

EXAMPLE 3 Verifying Triangle Congruence



Show that the triangles are congruent for the given value of the variable.

A $\triangle UVW \cong \triangle YXW, x = 3$

$$ZY = x - 1$$

$$= 3 - 1 = 2$$

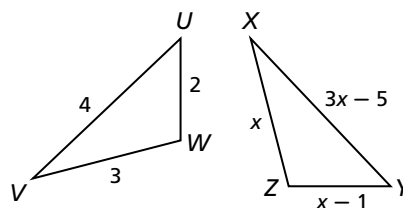
$$XZ = x = 3$$

$$XY = 3x - 5$$

$$= 3(3) - 5 = 4$$

$$\overline{UV} \cong \overline{YX}, \overline{VW} \cong \overline{XZ}, \text{ and } \overline{UW} \cong \overline{YZ}.$$

So $\triangle UVW \cong \triangle YXZ$ by SSS.



B $\triangle DEF \cong \triangle JGH, y = 7$

$$JG = 2y + 1$$

$$= 2(7) + 1$$

$$= 15$$

$$GH = y^2 - 4y + 3$$

$$= (7)^2 - 4(7) + 3$$

$$= 24$$

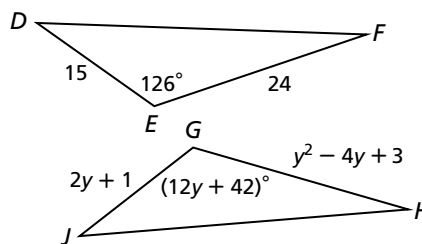
$$m\angle G = 12y + 42$$

$$= 12(7) + 42$$

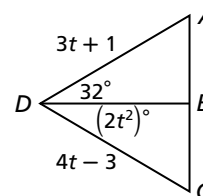
$$= 126^\circ$$

$$\overline{DE} \cong \overline{JG}, \overline{EF} \cong \overline{GH}, \text{ and } \angle E \cong \angle G.$$

So $\triangle DEF \cong \triangle JGH$ by SAS.



3. $\triangle ADB \cong \triangle CDB, t = 4$

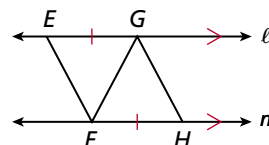


EXAMPLE 4 Proving Triangles Congruent

Given: $\ell \parallel m, \overline{EG} \cong \overline{HF}$

Prove: $\triangle EGF \cong \triangle HFG$

Proof:



Statements	Reasons
1. $\overline{EG} \cong \overline{HF}$	1. Given
2. $\ell \parallel m$	2. Given
3. $\angle EGF \cong \angle HFG$	3. Alt. Int. $\angle$ Thm.
4. $\overline{FG} \cong \overline{FG}$	4. Reflex Prop. of $\cong$
5. $\triangle EGF \cong \triangle HFG$	5. SAS Steps 1, 3, 4



4. Given: $\overrightarrow{QP}$ bisects $\angle RQS, \overline{QR} \cong \overline{QS}$

Prove: $\triangle RQP \cong \triangle SQP$

